deviation, Probability space, conditional probability, Baye's theorem, various types of distributions, uniform, binomial, poisson, normal and exponential distribution.

Section - 3 : Hydraulics and Water Resource Engineering

Fluid Mechanics: Properties of fluids, fluid statics; Continuity, momentum, and energy equations and their applications; Potential flow, Laminar, and turbulent flow; Flow in pipes, pipe networks; Concept of boundary layer and its growth; Concept of lift and drag. Open Channel Flow and Hydraulic Machines: Introduction of open channel flow, energy depth relationship, uniform flow, gradually varied flow, water surface profiles, and its computations, rapidly varied flow and its computations, spatially varied flow, unsteady flows, hydraulics of mobile bed channels, Fluid machinery; Turbines, pumps and hydro-electric power systems. Water Resources Engineering: Hydrologic cycle and its components, hydrograph analysis, flood estimation, and routing, surface runoff models, hydrological modelling, groundwater hydrology - steady state well hydraulics and aquifers; Application of Darcy's Law, Crop water requirements - Duty, delta, Diversion headworks, canal falls, Regulators modules, Design and construction of gravity dams, Theories of seepage and design of weirs, barrages and dams, spillways, energy dissipators.

Section - 4 : Structural Engineering

Mechanics of Materials: Bending moment and shear force in statically determinate beams; Simple stress and strain relationships; Theories of failures; Simple bending theory, flexural and shear stresses, shear centre; Uniform torsion, buckling of column, combined and direct bending stresses. Structural Analysis: Statically determinate and indeterminate structures by force/ energy methods; Method of superposition; Analysis of trusses, arches, beams, cables and frames; Displacement methods: Slope deflection and moment distribution methods; Influence lines; Stiffness and flexibility methods of structural analysis. Concrete Structures: Working stress, Limit state and ultimate load design concepts; Design of beams, slabs, columns, footing, staircase; Pre-stressed concrete.

Steel Structures: Working stress and Limit state design concepts; Design of tension and compression members, beams and beam-columns, column bases; Connections–simple and eccentric, beam-column connections, plate girders and trusses; Plastic analysis of beams and frames. Pre-stressed Concrete: Pre-stressing systems and end anchorages, losses of pre-stress. Analysis and design of members for flexure,shear,bond and bearings, Cable layouts, Design of Pre-stressed Bridges. Structural Dynamics: Free and Forced vibration analysis of Single and Multi degree of freedom systems. Orthogonality relationships of principal modes, Earthquake forces, nature and magnitude, pseudo-static method of approximate evaluation of earthquake forces, seismicity, Earthquake Motion and Response, Response Spectra, Philosophy of Capacity Design. Concepts of seismic design: Earthquake resistant design, construction & detailing for RCC and Masonry structures and relevant codes such as IS 1893:2002, IS 13920, IS:4326etc.

Section - 5 : Geotechnical Engineering

Soil Mechanics and Geotechnical Engineering- Nature of soil, soil formation and soil type, soil properties, basic definitions, phase relations, index properties, basic concepts of clay minerals and soil structure, soil classification and identification, hydraulic conductivity, seepage, compaction, stress distribution, shear strength, Mohr's circle of stress, Mohr-Coulomb failure theory, shear strength parameters, compressibility and consolidation, soil exploration, shallow foundations, settlements of footings andrafts, pile foundations. Geotechnical Earthquake Engineering - Engineering seismology,seismic risks and hazards, social and economic consequences, nature and attenuation of earthquake magnitude, ground motion, site characteristics, dynamic behavior of soils, liquefaction and cyclic mobility, seismic response of soil structure system, mitigation techniques. Rock Mechanics - Problems of rock mechanics, rock exploration, Griffith's theory, Coulomb's theory, in-situ tests on rock mass, mechanical, thermal and electrical properties of rock mass, pressure tunnels, lined and unlined tunnels, foundation on rocks, slope stability, rock bolt anchors and grouting, problems associated with tunnels, tunnelling in various subsoil conditions and rocks. Uncertainties, Risk and Reliability in Geotechnical Engineering-Risk,randomness, uncertainty, measures of reliability, modeling of uncertainty, probability, tests of goodness-of-fit (chi-square test, Kolmogorov-Smirnov test), reliability methods, deterministic and probabilistic approaches, Monte Carlo simulation and applications, risk assessment. Ground Improvement Techniques-Mechanical modification, precompression, sand drains, soil stabilisation, chemical modifications and grouting, hydraulic modification, ground modification by soil reinforcement, difficult soils.

5. Department of Computer Science and Engineering/ Department of Information Technology / Department of Software Engineering

Computing Related Mathematics: Propositional and first-order logic, sets, relations, functions, partial order and lattice groups. Groups. Vectors, matrices, determinants, System of linear equations, eigenvalues and eigenvectors, LU decomposition. Vector space, differential equation gradients, maxima minima random variables. Uniform Gaussian exponential, Poisson and binomial distributions. Mean Median Mode and standard deviation. Conditional Probability and Bayes Theorem.

Programming and Data Structures: Programming in C, recursion, arrays, stacks, queues, linked list, trees, binary heaps, graphs.

Algorithms: Searching sorting hashing asymptotic worst-case time and space complexity algorithm design techniques: Greedy, dynamic programming and divideand conquer. Graph search, minimum spanning trees, shortest path.

Theory of Computation: Regular expression and finite automata context-free grammar and pushdown automata, regular and context-free languages,pumping Lemma, Turing machines and undecidability.

Compiler Design: Lexical analysis parsing syntax-directed translation runtime environments intermediate code generation

Operating System: Processes inter-process communication concurrency and synchronisation, deadlock CPU scheduling, memory management and virtual memory file systems.

Database Management Systems: ER model, relational algebra, tuple calculus,SQ Lintegrity constraints, normal forms, file organisation, indexing (B and B plus trees) transactions and concurrency control.

Computer Networks: Concept of layering. LAN Technologies (Ethernet) flow and error control techniques, switching ipv4/ ipv5 routers and routing algorithms (distance vector, link state). TCP/UDP and socket congestion control, Application layer protocols (DNS, SMTP POP3 HTTP), Basics of Wi-Fi, network security: Authentication, basics of Public key and Private key cryptography digital signatures and certificates, firewalls.

CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT), and Email.

Software Engineering: Software development life cycle, requirement and feasibility analysis, data flow diagrams, process specifications, input/ output planning and managing the project, design, coding, software testing, implementation, maintenance, software metrics.

Web Technologies: Web IR retrieval,Web mining, Bigdata, NOSQL, Basics of cloud (SaaS, PaaS, IaaS, Public and Private Cloud).

Artificial Intelligence: Artificial intelligence approach for problem-solving, Automated reasoning for prepositional logic, state-space representation of problems, bounding functions, breadth-first search, depth- first search, A, A*, Ao*, etc. Frames scripts semantic nets, production system, procedural representation.